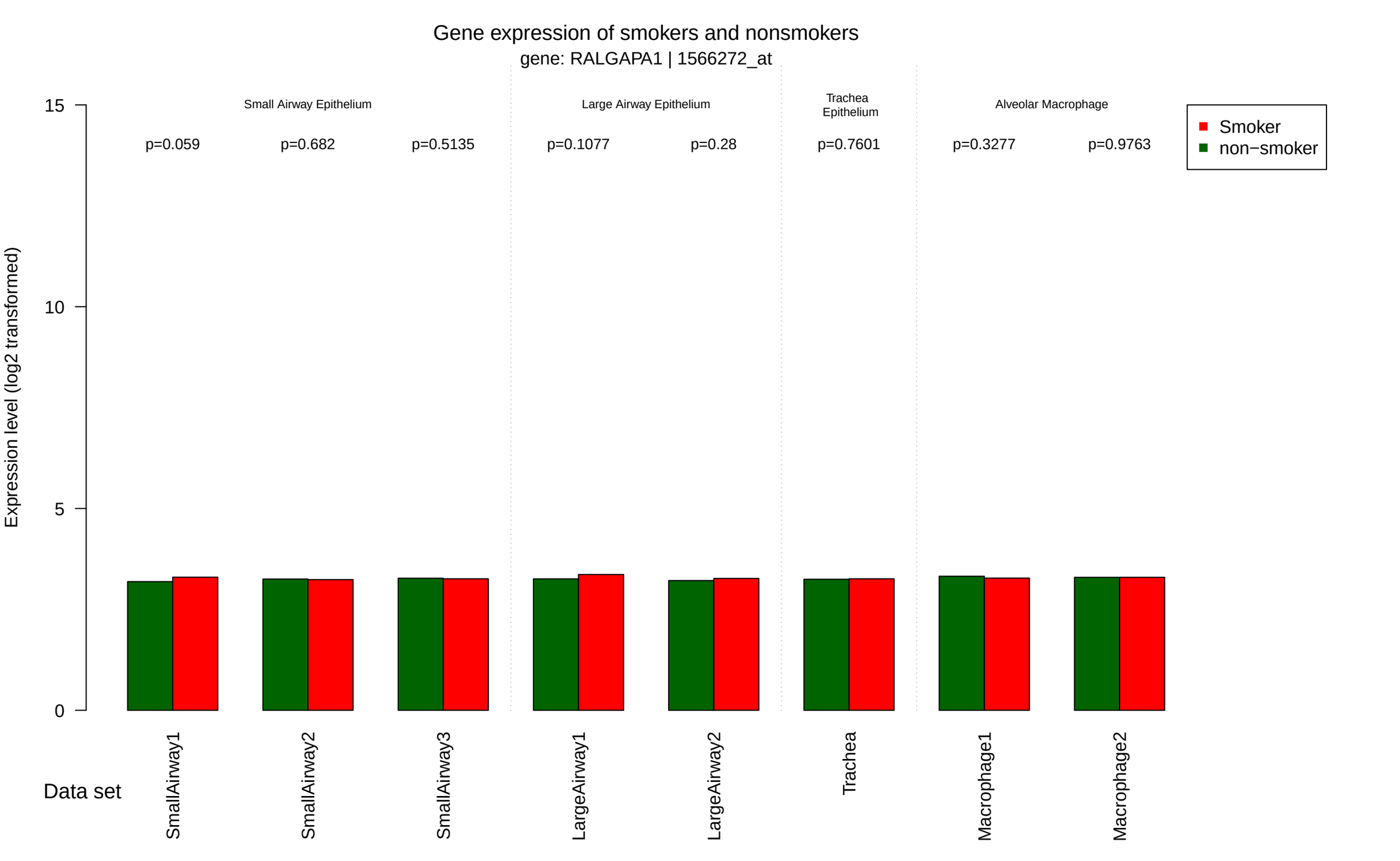
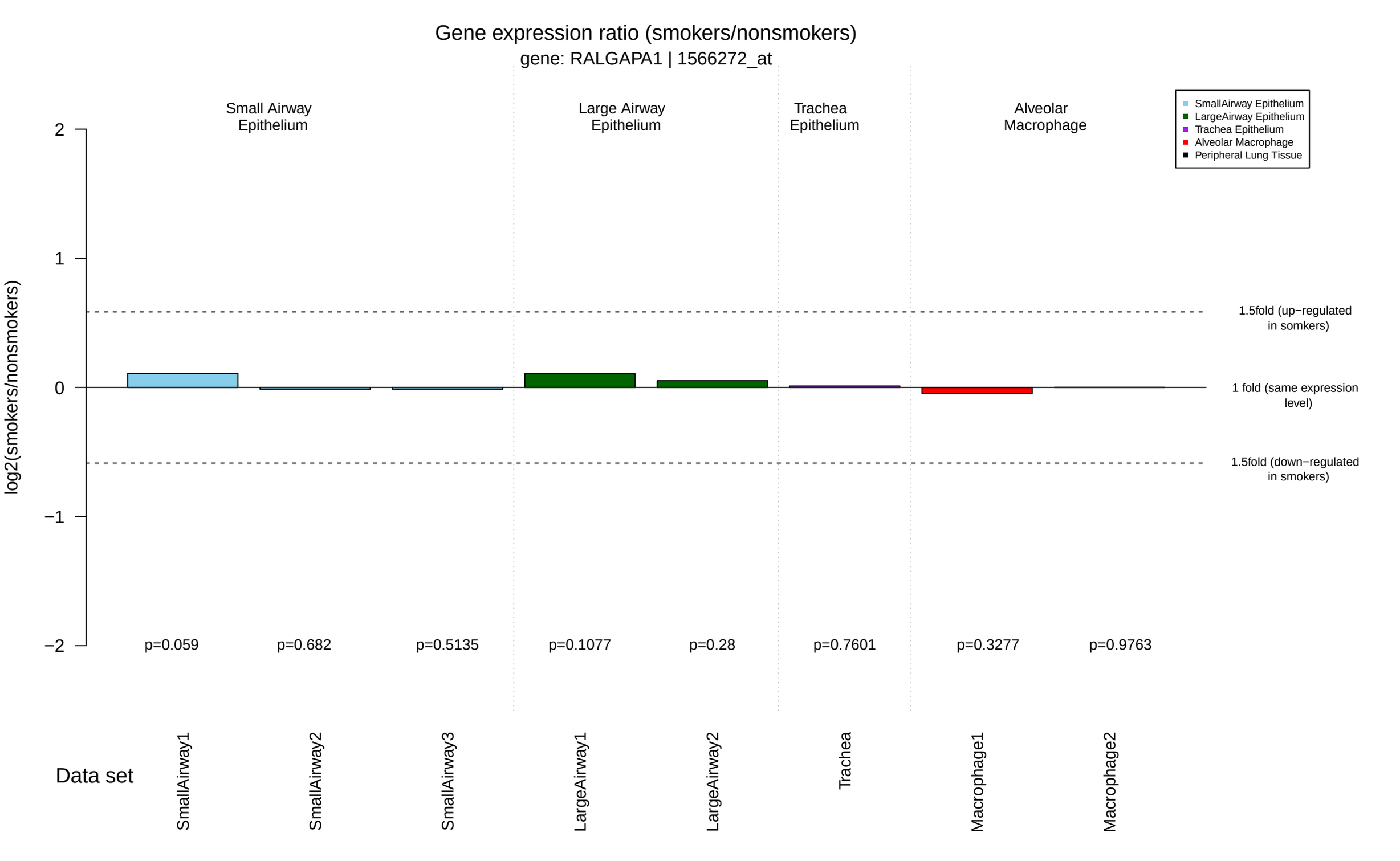
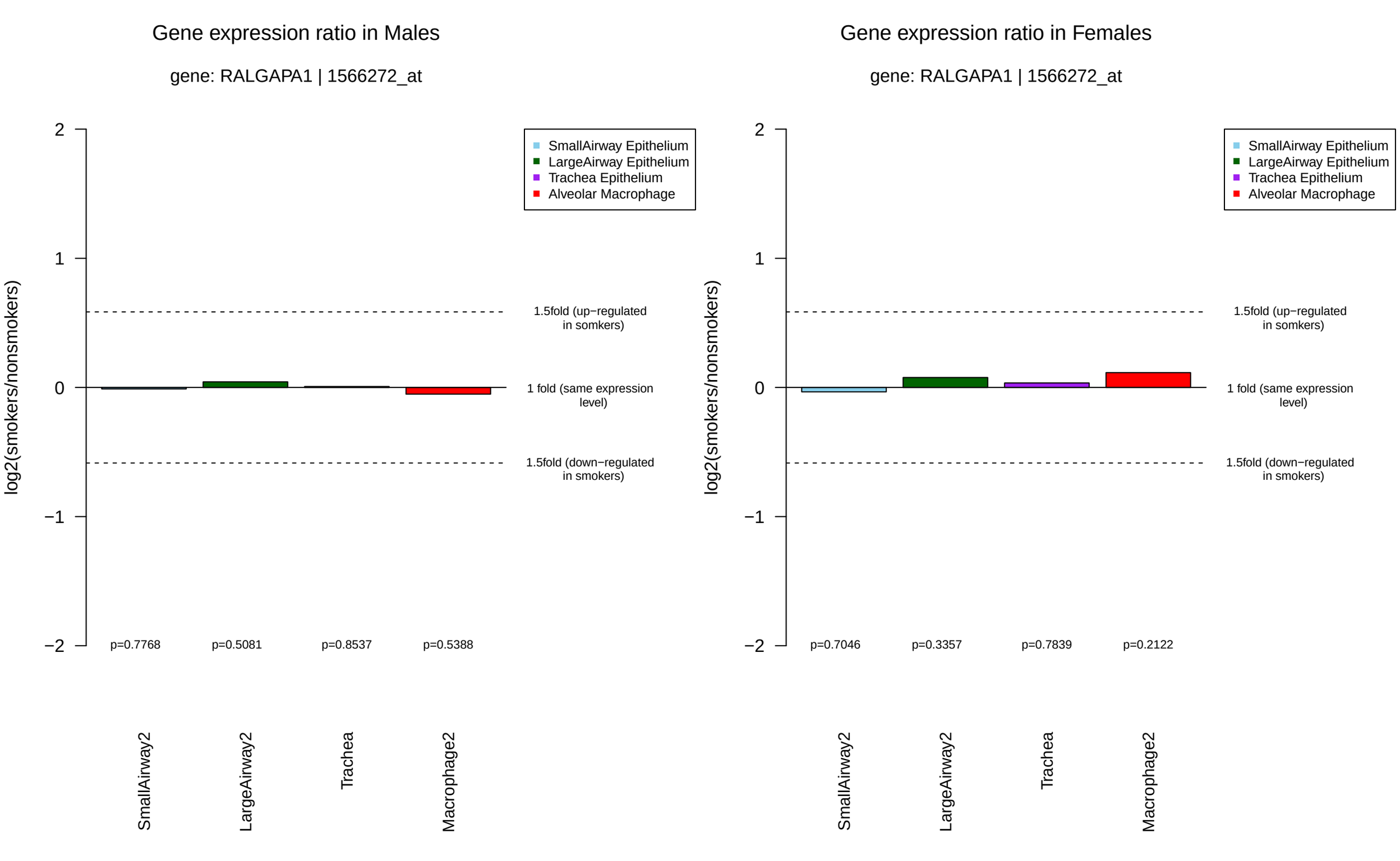
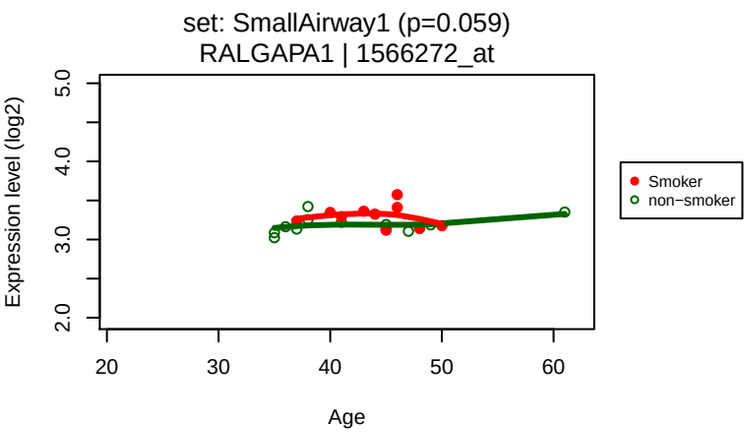
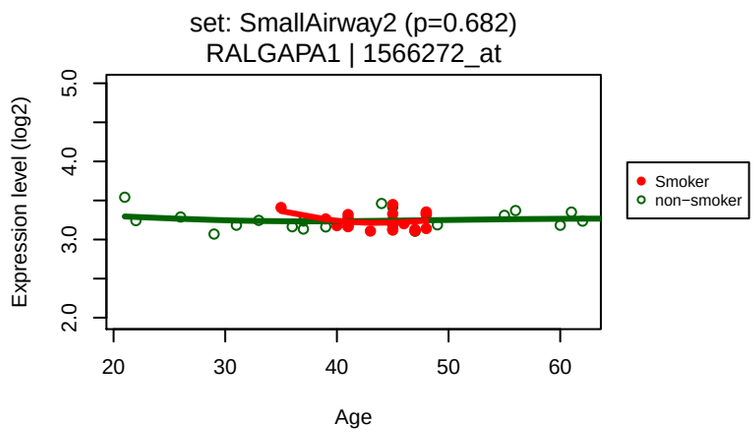
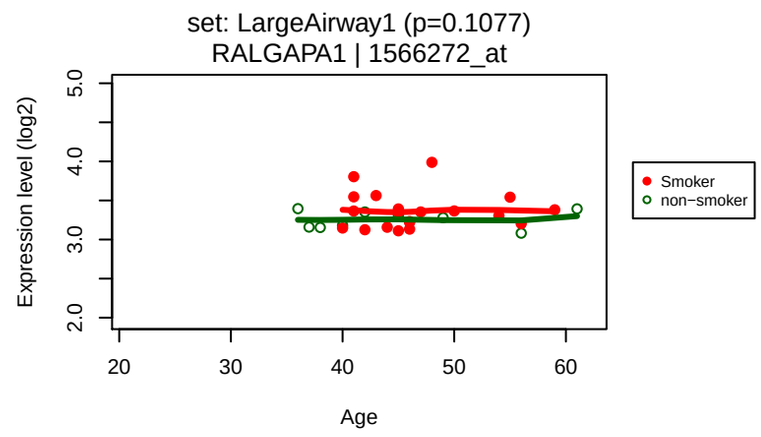

Gene expression of smokers and nonsmokers
gene: RALGAPA1 | 1566272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.059
p=0.682
p=0.5135
p=0.1077
p=0.28
p=0.7601
p=0.3277
p=0.9763
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

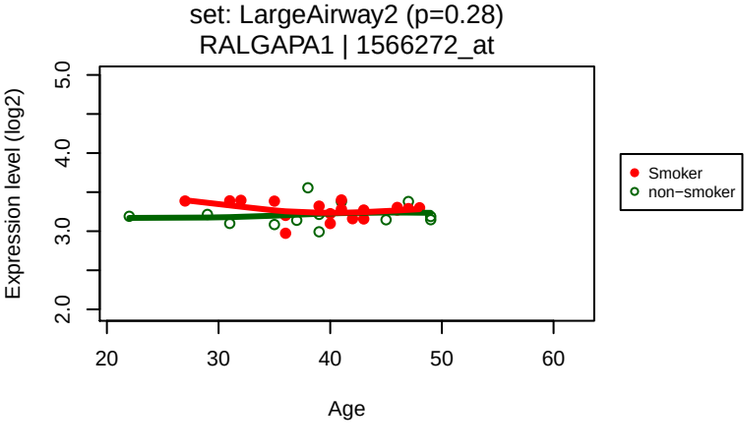

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA1 | 1566272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.059
p=0.682
p=0.5135
p=0.1077
p=0.28
p=0.7601
p=0.3277
p=0.9763
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

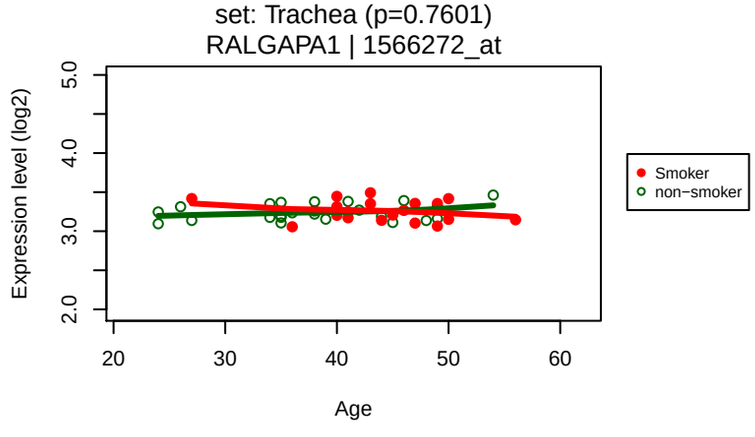

Gene expression ratio in Males
gene: RALGAPA1 | 1566272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7768
p=0.5081
p=0.8537
p=0.5388
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA1 | 1566272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7046
p=0.3357
p=0.7839
p=0.2122
SmallAirway2
LargeAirway2
Trachea
Macrophage2

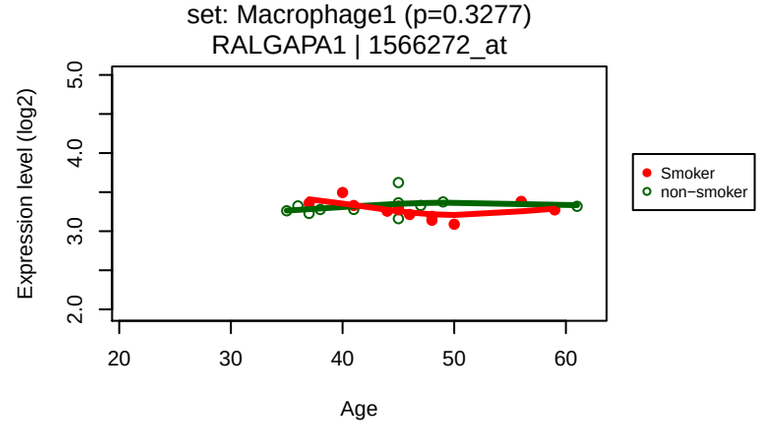

set: SmallAirway1 (p=0.059)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker

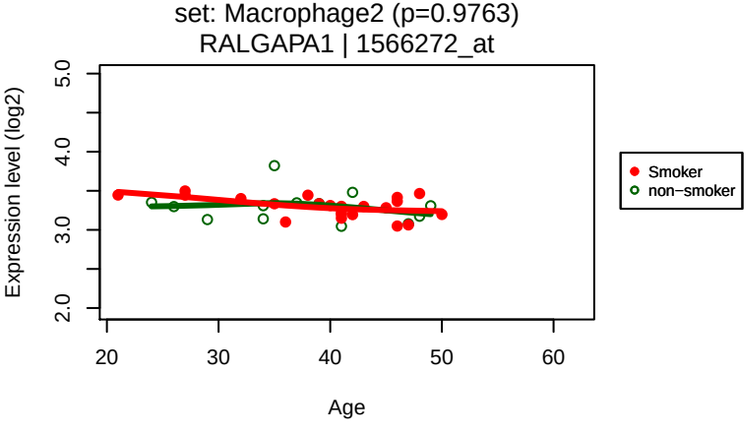

set: SmallAirway2 (p=0.682)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker

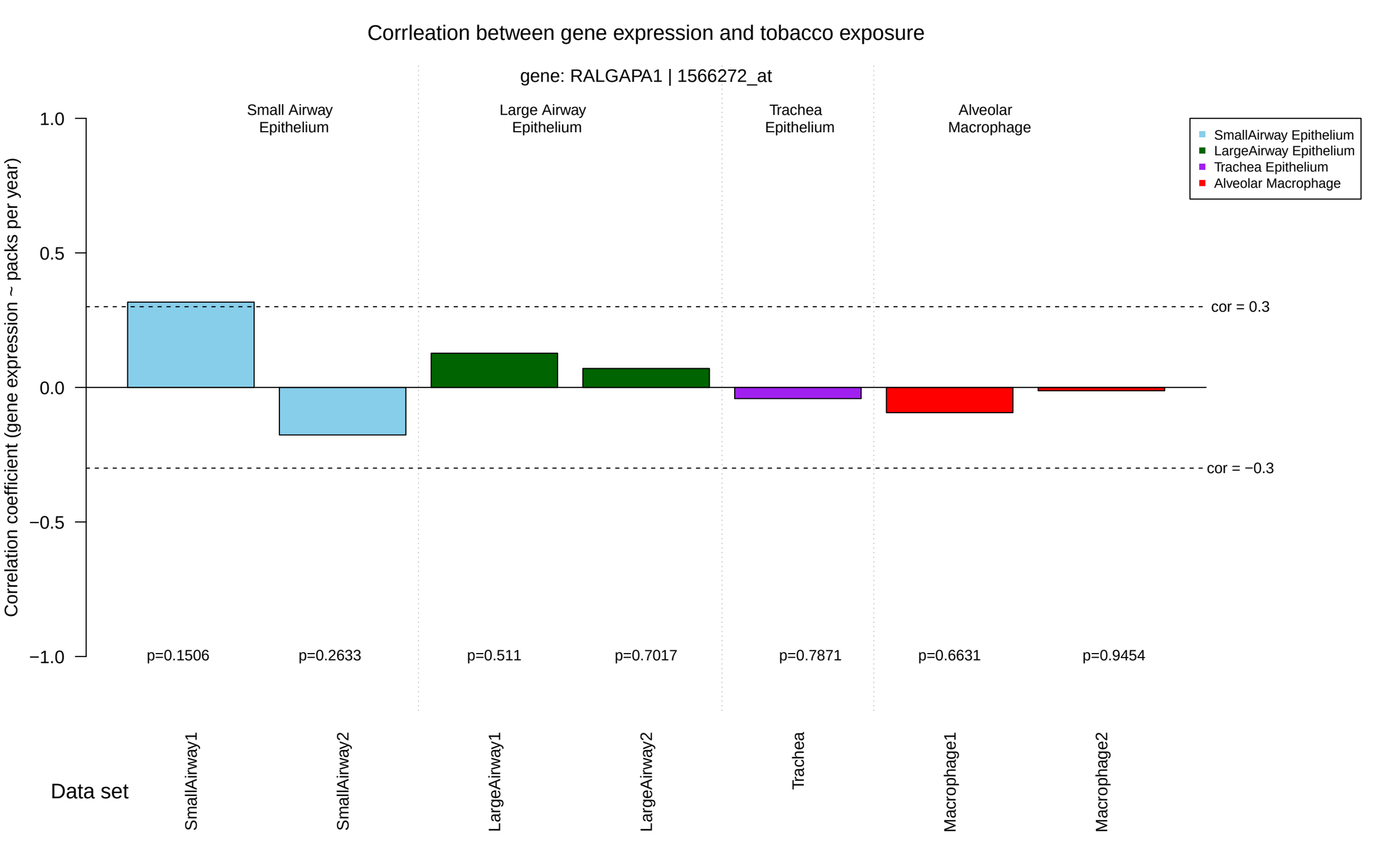

set: LargeAirway1 (p=0.1077)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.28)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7601)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3277)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9763)
RALGAPA1 | 1566272_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA1 | 1566272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1506
p=0.2633
p=0.511
p=0.7017
p=0.7871
p=0.6631
p=0.9454
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2